

Lavneet Sidhu

 Lavneet Sidhu |  Lavneet.s.sidhu@gmail.com |  6476075932

EDUCATION

Bachelor of Applied Science in Computer Engineering

September 2023 - June 2028

University of Toronto, Toronto, Ontario

- **Minors:** Artificial Intelligence, Robotics, Business
- **Relevant Courses:** Robot Modeling & Control (ECE470), Applied Fundamentals of Deep Learning (APS360), Introduction to Control Systems (ECE311), Operating Systems (ECE344), Algorithms and Data Structures (ECE345)

PROFESSIONAL EXPERIENCE

Co-Founder, Engineering

2026 - Present

Project Heimdall (Pre-Seed Startup)

- Co-founded a pre-seed counter-UAS startup fusing **radar**, **RF**, **acoustic**, and **EO/IR** sensor data into a single real-time air picture; still in active technical development, having completed an interview with **Creative Destruction Lab (CDL)** for its mentorship stream.
- Architected the multi-sensor **fusion and tracking engine** in Python, implementing gated **global-nearest-neighbour association** and per-track **Kalman filtering** across a full tentative-confirmed-coasted track lifecycle.
- Built a deterministic, seeded **simulation benchmark** achieving **98.9% track completeness** with zero ID switches at 3.8m RMSE, and 86.3% completeness in a complex multi-target airport scenario.

Firmware Developer Intern

May 2026 - May 2027

Qualcomm

- Engineered virtual platforms for pre-silicon CPU and SoC modelling using **C/C++** and **SystemC**, optimizing hardware architecture evaluation for next-generation **CPUs**.
- Designed and integrated **transaction-level models (TLM)** to simulate processor architectures, bridging hardware simulation and real-time firmware execution for complex autonomous systems.
- Set up **CI/CD** pipelines and developed automated regression tests using **Python**, **Docker**, and **Git**, ensuring cycle-approximate accuracy against RTL reference designs for system determinism.

R&D Team Lead, Autonomous Rover Tracking

September 2025 - Present

University of Toronto – Research under Professor Sinisa Colic

- Leading a team of 6 engineers building a computer vision-based rover tracking platform for autonomous navigation **R&D**.
- Architected a **sensor-fusion** pipeline combining camera, LiDAR, and proximity sensors into a rover state vector.
- Integrated hardware sensing units with cloud-based storage and an interactive dashboard for post-processing analytics.

Machine Learning & LLM Researcher

May 2025 - Present

University of Toronto, Dalla Lana School of Public Health

- Optimized sentiment analysis and labeling pipelines by fine-tuning **LLMs** and traditional **NLP transformers**.
- Investigated and implemented **novel prompt engineering strategies** to significantly enhance LLM inference performance, with direct applications in large-scale sentiment analysis and programmatic tasks.
- Currently **authoring a comprehensive research paper** alongside Dr. Jude Kong at the Dalla Lana School of Public Health detailing recent artificial intelligence advancements for large-scale programmatic analysis.

PROJECTS & AWARDS

Autonomous Drone Software – UTAT

Sept 2025 - May 2026

- Built autonomous aerial software using **ROS2**, **PX4**, and **Gazebo**, with a **YOLOv8/OpenCV** vision pipeline and **LiDAR** integration on an NVIDIA Jetson Orin for competitive autonomous aerial missions.

1st Place Canadian Engineering Competition (Programming)

2025, 2026

- Swept 1st at the university (UTEK), provincial (OEC), and national (CEC) levels, topping 55 Canadian universities for U of T's first programming win in two decades.
- Built the winning 2026 project, an **Augmented Reality inspection platform** that lets inspectors log and annotate defects in 3D through a **Unity** frontend, backed by a locally emulated **DynamoDB** server, with real-time data sharing to stakeholders via a **React** dashboard.

TECHNICAL SKILLS

- **AI/ML & Cloud:** PyTorch, TensorFlow, AWS, LLM Prompt Engineering, NLP, OpenCV, YOLOv8, ROS2
- **Embedded & Real-Time Systems:** RTOS, Real-Time DSP, Closed-Loop Control, Kalman Filtering & Sensor Fusion, FPGA/RTL (Verilog), PCB Design (Altium), Linux Driver Development, SystemC/TLM
- **Programming Languages:** Python, C, C++, Rust, Verilog, SQL, R, Java, JavaScript, Node.js, React, MATLAB